

Topological Game-Based Autonomous Driving: A Road Network Self-Play Decision Framework Based on AlphaZero Paradigm

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1 Abstract

Current mainstream training paradigms for autonomous driving decision-making heavily rely on massive real-world driving data and high-performance computing clusters. Typical approaches include end-to-end visual imitation learning and manually customized scenario-based reinforcement learning, which suffer from poor model interpretability, insufficient coverage of edge cases, and weak generalization to unseen road topologies. Such data- and computation-intensive development paths are difficult to replicate under domestic industrial constraints. To address these challenges, this paper proposes a **topological game-based autonomous driving paradigm**. Inspired by the self-play iterative mechanism of AlphaZero in board games, the proposed framework decomposes complex urban road networks into enumerable basic road primitives, constructing programmatically generable, topologically complete, resettable micro road game boards. The vehicle driving decision problem is reformulated from a conventional continuous physical control task into a **temporal multi-objective combinatorial game problem defined on structured road topologies**.

This framework abandons brute-force fitting on pixel-level data. It adopts graph neural networks to realize structured topological embedding of road networks, leverages Monte-Carlo Tree Search (MCTS) for multi-step lookahead optimal decision reasoning, and balances safety, compliance, efficiency and comfort via a hierarchical multi-objective reward system. High-quality driving trajectories are autonomously generated through simulation self-play, and a search distillation strategy is adopted to achieve lightweight deployment on vehicle-end devices. Furthermore, this paradigm is extended from driving topologies to parking topologies, with dedicated parking primitives, fine-grained discrete action sets and pose optimization reward functions added. A unified game decision mechanism for **driving-parking cross-scene tasks** is realized within a single network architecture, verifying the general modeling capability of the

proposed framework for cross-scale, full-domain autonomous driving tasks.

Belonging to the **Learning-based Motion Planning** paradigm, the proposed framework does not rely on manually enumerated interactive scenarios, and complex interactive driving behaviors emerge autonomously driven by topological priors and self-play mechanisms. Theoretical analysis and experimental logic show that the proposed paradigm compresses high-dimensional state space via structural road priors, significantly reducing the dependence on human-labeled driving data and high-end computing resources. It possesses five core advantages: interpretable decision-making, topologically complete scenario coverage, cross-road generalization, autonomous emergence of interactive driving behaviors, and native integrated driving-parking modeling. This work provides a novel formal academic framework for the training, verification and generalization of autonomous driving decision models, serving as a differentiated intelligent driving technical path suitable for domestic constraints on data and computing resources.

Keywords: Autonomous Driving; Topological Game; Road Network Primitives; Integrated Driving and Parking; Motion Planning; AlphaZero; Monte-Carlo Tree Search; Self-Play Reinforcement Learning; Graph Neural Networks

2 Introduction

2.1 Core Bottlenecks of Existing Autonomous Driving Paradigms

At present, intelligent decision systems for autonomous driving are mainly divided into two mainstream technical routes: **end-to-end imitation learning** and **scenario-based reinforcement learning**. Although both have achieved phased results in mass production, their inherent paradigm defects increasingly hinder the breakthrough toward general autonomous driving.

The **end-to-end pure visual imitation learning paradigm**, represented by Tesla FSD, relies on millions of fleet-scale closed-loop data and dedicated supercomputing clusters to fit human driving behaviors from massive real-world trajectories. Essentially a statistical fitting task in high-dimensional pixel space, this paradigm has three inherent drawbacks. First, it has a **strong dependence on data and computing resources**, and the performance improvement relies entirely on mileage accumulation and computing iteration, which is difficult to replicate for domestic industries without equivalent data and computing scale. Second, the model is a **black-box system** without explicit logical support for decision-making, making it impossible to trace failure causes and failing to meet the safety interpretability requirements of high-level autonomous driving. Third, the model lacks **topological generalization ability**, as it only learns visual pixel features rather than the connectivity of road topologies, leading to decision failures when encountering unfamiliar urban road layouts.

The **manual scenario-based reinforcement learning paradigm**, implemented on simulation platforms such as CARLA and LGSVL, trains models via manually constructed scenarios including intersections, highways and urban areas. Its core weakness is the **incompleteness of scenario coverage**: all test scenarios are manually customized and cannot systematically cover the topological combinations of global road

networks. Edge cases (Corner Cases) rely on random road testing sampling, which cannot guarantee the robustness and generalization of decision models from a structural perspective, resulting in easy overfitting to fixed scenarios.

Traditional **rule-based and sampling-based motion planning methods** (A^* , RRT, Lattice) rely on manually designed cost functions and path templates to generate deterministic collision avoidance trajectories. However, they **cannot model social driving interactions such as multi-vehicle game negotiation, yielding and lane merging**, and struggle to cope with dynamic, game-like real-world road scenarios.

In addition, most industry-level integrated driving-parking solutions are essentially **hardware and software function stacking**, where driving and parking decision-making belong to two independent logical systems without a unified spatial topological modeling theory, failing to realize native general decision-making at the algorithm level.

Fundamentally, existing paradigms **ignore the structural nature of road networks**: urban road networks are not random unstructured scenes, but combinatorial systems assembled, nested and deformed from a limited set of basic primitives. Conventional methods define driving as a pure physical motion control problem and abandon road topological priors, leading to infinite explosion of state space, low learning efficiency, poor generalization, and the inability to uniformly address macro road traversal and micro parking-space occupancy game problems.

2.2 Feasibility of Migrating the AlphaZero Game Paradigm

AlphaZero achieved a revolutionary breakthrough in board games by realizing **human-prior-free, rule-driven self-play emergence of optimal strategies**. Its core architecture consists of four key components: a structured closed game environment, discrete sequential decision actions, policy-value dual-network evaluation, multi-step lookahead via Monte-Carlo Tree Search, and self-iterative data generation. This paradigm completely breaks away from the dependence on human expert data and converges to optimal strategies through constrained autonomous exploration in complex game scenarios.

Driving scenarios share an underlying logical homology with board games: road networks have fixed topological structures and traffic rules, and driving behaviors are sequential, constrained decision-making processes, which fully satisfy the game modeling conditions of **closed rules, finite actions, sequential iteration and quantifiable rewards**. Based on this, this paper migrates the AlphaZero game paradigm to autonomous driving: **constructing game boards via road topologies, replacing board moves with driving decisions, defining game constraints via traffic rules, and generating driving strategies via self-play**, completely breaking out of the data-fitting development path.

Meanwhile, both driving and parking can be uniformly abstracted as **topological space occupancy games with different scales and constraints**, which naturally fits the structured modeling logic of the proposed paradigm and provides theoretical feasibility for unified full-domain decision-making.

2.3 Core Innovations and Contributions of This Work

1. **Paradigm Innovation:** A topological game-based autonomous driving framework is first proposed, transforming the physical driving control problem into a road network topological combinatorial game problem, establishing a novel motion planning modeling system of “road primitives - sequential moves - multi-objective game”.
2. **Scenario Innovation:** Seven core road primitives are defined to generate complete driving scenarios, and an extended parking topological primitive is newly added to realize unified topological modeling of macro roads and micro parking lots.
3. **Algorithm Innovation:** A complete algorithm pipeline including **topological GNN encoding, multi-objective MCTS-UCT search, self-play iteration and policy distillation** is designed to realize simulation self-play training without human driving data, enabling the autonomous emergence of social interactive behaviors such as intersection yielding, ramp merging and parking adjustment.
4. **Engineering Innovation:** A hierarchical architecture of “MCTS optimal search + network distillation lightweighting” is constructed to realize **native integrated driving-parking decision-making at the algorithm level**, rather than traditional functional stacking schemes.
5. **Theoretical Innovation:** A formally verifiable training and verification framework for intelligent driving is built, providing demonstrable theoretical support for the generalization, safety and cross-scene universality of autonomous driving decisions.

3 Related Work

3.1 Geometric Deep Learning for Road Topology Modeling

Geometric deep learning (GDL) has become the dominant paradigm for modeling unstructured and topological spatial data, enabling permutation-invariant and structure-aware feature learning on graphs and manifolds. In autonomous driving, existing GDL-based methods primarily focus on road graph representation, traffic flow prediction, and vehicle trajectory forecasting. By encoding lane connections, intersection topology, and road contextual attributes, graph neural networks (GNNs) significantly outperform conventional CNN-based methods in generalization across unseen road layouts.

However, current GDL applications in driving remain confined to **static topological perception and prediction tasks**. Most existing works treat road topology as a fixed environment feature and learn implicit spatial patterns for downstream forecasting, lacking explicit modeling of **sequential decision-making, multi-agent game interaction, and long-term planning**. Furthermore, few frameworks unify macro-scale

road topology and micro-scale parking topology within a single geometric embedding space, resulting in separate models for driving and parking scenarios.

Different from existing GDL pipelines, this work elevates geometric topology representation from passive perception to **active topological game decision-making**. The road graph is constructed as a programmable game board, and driving/parking behaviors are modeled as sequential topological moves, enabling GDL-based structural prior to support iterative self-play policy optimization rather than only static feature extraction.

3.2 Topology-Aware Mixture-of-Experts for Multi-Scale Graph Learning

Topology-aware Mixture-of-Experts (TopoMoE) architectures introduce sparse expert routing conditioned on structural graph properties, enabling adaptive feature extraction for heterogeneous topological subgraphs. Compared with fixed-structure GNNs, TopoMoE dynamically assigns different network experts to process graph regions with distinct density, connectivity, and structural patterns, achieving superior scalability and representation diversity.

In existing robotic and autonomous driving literature, TopoMoE is mainly adopted for scene classification, graph segmentation, and environment adaptive perception. Nevertheless, existing TopoMoE designs fail to address the **cross-scale topological heterogeneity** between sparse road-level traffic scenarios and dense parking-field scenarios, and are rarely deployed for motion planning and sequential decision tasks.

In this framework, we customize a decision-oriented TopoMoE backbone to unify dual-scale topological modeling. Macro sparse road topologies and micro dense parking topologies are automatically routed to corresponding experts, realizing **native driving-parking unified encoding** without two-branch structural stacking. This provides a critical intermediate layer for cross-scene generalization in topological game planning.

3.3 Topological Reinforcement Learning and GRPO Optimization

Traditional reinforcement learning (RL) methods for autonomous driving, such as PPO and A2C, suffer from high training variance, unstable multi-agent credit assignment, and over-conservative convergence in interactive traffic scenarios. To address these issues, group relative policy optimization (GRPO) eliminates the independent critic network and optimizes policy advantages via intra-group trajectory ranking, significantly reducing computational overhead and improving training stability.

Topology-aware GRPO further introduces graph topological constraints into group sampling and advantage calculation, making RL updates aware of spatial adjacency, structural connectivity, and regional topological differences. Despite its advantages, existing topological GRPO research is limited to multi-agent communication topology optimization and simple grid-world control, and has not been applied to **long-horizon Monte-Carlo tree search (MCTS) self-play motion planning**.

Different from prior topological RL works, this framework integrates Topology-GRPO into an AlphaZero-style self-play iterative pipeline. MCTS generates high-quality topological game trajectories as teacher supervision, and GRPO stabilizes pol-

icy update via relative group advantages, solving the easy-collapse and high-variance problems inherent in traffic game self-play.

3.4 Differences and Superiority of the Proposed Framework

In summary, existing geometric learning, TopoMoE, and topological RL methods only solve **separate sub-problems** of topological perception, multi-scale encoding, and policy optimization. No existing work has systematically integrated these three layers to construct a **unified topological game decision paradigm** for full-domain autonomous driving.

This framework establishes a complete three-layer technical system:

1. **GDL provides fundamental geometric topological prior and structured state representation;**
2. **TopoMoE realizes adaptive multi-scale unification of driving and parking topology;**
3. **Topology-GRPO stabilizes MCTS self-play iteration and topological game policy optimization.**

Compared with previous learning-based motion planning methods, our paradigm transforms autonomous driving from data-intensive imitation or rule-dependent deterministic planning into **topology-structured, self-play-evolving, geometry-consistent game planning**, achieving better interpretability, structural generalization, and cross-scene universality.

4 Topological Game Modeling System for Autonomous Driving

4.1 Road Network Primitives: Basic Building Blocks of Game Boards

By decomposing the topology of urban roads, highways and rural road networks worldwide, seven core driving road primitives are abstracted as the basic units of the proposed paradigm. All real public road networks are combinations, stretches and nesting of these seven units:

1. Straight road unit: Basic linear structure for constant-speed car-following and straight traversal;
2. Orthogonal cross intersection unit: Standard four-way urban intersection topology covering straight crossing, conflicting traffic and turning game scenarios;
3. T-junction unit: Three-way branching topology adapted to branch merging into main roads and lateral intersection traversal;
4. Roundabout unit: Closed circular topology with multiple radial entrances, representing typical unordered multi-vehicle game scenarios;

5. Fishbone strip unit: Trunk-branch strip topology adapted to long and narrow urban roads along rivers and traffic arteries;
6. Radial hub unit: Central-node radial topology corresponding to radial road structures in old urban cores;
7. Branching tree unit: Hierarchical branching topology without closed loops, adapted to suburban, mountainous and rural unstructured road networks.

To further expand the universality of the paradigm, this paper adds an eighth extended primitive: **Parking Lot Topology Unit**, as a general extended scenario for the integrated driving-parking capability of the framework, modeling closed, high-density, highly constrained micro-field games:

4.1.1 Parking Lot Topology Unit (Extended General Unit)

Different from macro public road topologies, parking lots belong to **high-density, narrow-channel structured topologies with termination constraints**, which can be uniformly incorporated into the graph topology system of this framework:

- Topology Definition: Grid topology composed of main access edges, secondary channel edges and parking-space termination nodes;
- Parking Space Modeling: Vertical, parallel and angled parking spaces are uniformly abstracted as **terminating topological branches with boundary constraints**;
- Field Characteristics: Narrow passages, high pose accuracy requirements and compact traversable space;
- Game Characteristics: Multi-vehicle parking competition, channel passing, multiple adjustment maneuvers and safe distance games, belonging to typical high-density, long-sequence decision games;
- Scale Characteristics: Defined as **micro topological games**, forming a full-domain coverage of “large-scale road networks - small-scale parking fields” together with macro driving topologies.

Essentially, driving road networks and parking lots are **structured topological spaces with different scales and densities**. The proposed framework adaptively adjusts receptive fields via multi-scale graph convolutions to realize unified feature extraction for both macro road networks and micro parking fields, which is the core foundation for realizing algorithm-native integrated driving and parking.

Based on the above eight primitives, the framework can programmatically generate **macro driving game boards, micro parking game boards, and transition scenario boards connecting driving and parking**, truly achieving topologically complete coverage of full autonomous driving scenarios.

The core paradigm of this study takes the first seven driving primitives as the main body and the eighth parking primitive as a general extension, proving that the topological game idea is not limited to road driving, but **applicable to all structured and gameable traffic spaces**.

4.2 Core Mapping Relationship of the Game System

Strictly aligned with the AlphaZero board game logic, a unified game mapping system for driving and parking is established:

AlphaZero Board Game Concept	Unified Definition in Topological Game Autonomous Driving
Fixed Game Board	Multi-scale topological game fields programmatically generated by road/parking p
Discrete Moves	Macro driving decision moves + fine-grained sequential parking moves
Game Rules	Traffic regulations, vehicle dynamics, spatial boundaries and parking constraints
Game Reward	Safe and efficient driving / accurate collision-free parking
Expert Game Records	Sequential trajectories generated by full-scenario self-play
Board Game Patterns	Topological features of road primitives and parking primitives

4.3 Structured Representation of Game States

Abandoning high-dimensional raw image input, **graph-structured state representation** is adopted to define the global game state, greatly compressing the state space and ensuring topological generalization. The global state is defined as:

$$S = (G_{road}, Ego_{state}, Traffic_{state}, Light_{state})$$

where:

- G_{road} denotes the multi-scale topological attribute graph (unified graph structure for macro road networks and micro parking lots);
- Ego_{state} denotes the ego-vehicle motion and pose state;
- $Traffic_{state}$ denotes the dynamic traffic/field obstacle state;
- $Light_{state}$ denotes road rules and field boundary constraints.

4.4 Discrete Sequential Driving Action Space

Based on the unified move idea, a two-layer action system of **basic driving action set + fine-grained parking action set** is constructed to ensure paradigm consistency while adapting to the characteristics of both scenarios.

4.4.1 Basic Driving Action Set (Core)

It includes 6 types of longitudinal speed moves and 7 types of lateral lane moves, combining into 42 legal high-level driving decisions. An illegal action mask is configured for hard rule constraints, and high-level decision intentions are smoothly output via underlying continuous control.

4.4.2 Fine-Grained Parking Action Set (Extension)

Aiming at the characteristics of parking scenarios including **small displacement, multiple iterations, large steering angles and sequential adjustment maneuvers**, a series of P-type fine-grained moves are added to adapt to micro topological games:

- P0: Centimeter-level forward fine-tuning
- P1: Centimeter-level backward fine-tuning
- P2: Large-angle steering entry action
- P3: In-situ / small-range parking adjustment action

In parking scenarios, MCTS automatically **deepens the search depth and extends the sequential planning length**, giving full play to the natural advantages of AlphaZero in long-sequence multi-step games, which perfectly adapts to the multi-step combinatorial decision characteristics of parking maneuvers.

4.4.3 Unified Constraint Mechanism and Explanation of Action Hierarchy

For driving scenarios, actions such as entering intersections under red lights and solid-line lane changes are prohibited. For parking scenarios, actions exceeding boundaries, crossing parking lines and occupying obstacle areas are prohibited. An Action Mask is used throughout to block illegal moves.

Key Explanation: The discrete sequential moves in this study are **abstractions at the high-level behavioral decision layer**, rather than underlying continuous motion control outputs. This design adapts to the sequential reasoning logic of MCTS tree search, instead of directly commanding vehicles to execute step-like discrete motions. All high-level move decisions are finally smoothed via continuous trajectory interpolation and model predictive control to ensure driving continuity and ride comfort.

4.5 Hierarchical Multi-Objective Reward Function System

On the basis of the original hierarchical driving rewards, a dedicated parking reward branch is newly added to realize differentiated optimal trade-offs for the two scenarios.

4.5.1 Driving Rewards (Core)

Four-level trade-offs including safety hard constraints, compliance constraints, traffic efficiency and ride comfort.

4.5.2 Dedicated Parking Rewards (Extension)

1. **Pose Accuracy Reward:** Positive reward for minimizing parallelism deviation and center offset between the vehicle body and the parking space;
2. **Boundary Safety Reward:** Positive reward for maintaining safe distances from walls, adjacent vehicles and pillars;

3. **Parking Success Reward:** High efficiency reward for one-time accurate parking without repeated adjustments;
4. **Smoothness Constraint Reward:** Suppress frequent large steering wheel shaking and ineffective parking adjustments.

A hierarchical weighted reward scheme is adopted for basic paradigm verification with manually set hierarchical weights. Future work will conduct **reward weight sensitivity analysis experiments** and introduce **Inverse Reinforcement Learning (IRL)** to calibrate reward functions and reduce the subjectivity of manual parameter tuning.

5 Overall Architecture and Core Algorithm Design of Topological Game Autonomous Driving

5.1 Overview of the Five-Layer Overall Architecture

1. Road Network Topology Generation Layer: Generate multi-scale game road networks via primitive assembly;
2. Topology-Aware Encoding Layer: Embed features of road network topology graphs via GNN;
3. Topological Game Decision Layer: **GNN feature input** $\rightarrow$ **MCTS-UCT search** $\rightarrow$ **Policy/Value Network output** (core algorithm);
4. Policy Distillation Lightweight Layer: Distill MCTS teacher policies into vehicle-end lightweight planning networks;
5. Sim2Real Simulation Transfer Layer: Output decision-layer trajectories to connect with the perception and control modules of real vehicles.

The overall technical link supports a complete decision closed loop relying on a three-layer progressive topological learning system: **Geometric Deep Learning** constructs non-Euclidean topological representation priors to guarantee combinatorial generalization of modular road networks; **TopoMoE Topological Mixture of Experts** adaptively matches sparse macro driving topologies and dense micro parking topologies to realize unified driving-parking encoding; **Topology GRPO Topological Group Relative Policy Optimization** stabilizes MCTS self-play iteration, suppresses training variance and extreme Nash equilibrium convergence in multi-agent games, and finally realizes interpretable, generalizable and full-domain unified learning-based motion planning on structured road networks.

5.2 Topology-Aware Encoding Algorithm (GNN Topological Embedding)

5.2.1 Graph Structure Definition

The road network topology graph is defined as $G = (V, E, \mathcal{X}_V, \mathcal{X}_E)$

- V : Node set (intersections, road segment endpoints, parking-space termination nodes);
- E : Edge set (lane segments, access channels, parking-space branches);
- $\mathcal{X}_V$: Node features (node type, intersection priority, traffic light status, congestion occupancy);
- $\mathcal{X}_E$: Edge features (road grade, lane count, speed limit, solid/dashed lines, merging segment attributes).

5.2.2 Multi-Scale Graph Convolutional Encoding

Graph convolutional networks are adopted to aggregate local topological subgraphs:

$$h_v^{(l)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_v^{(l-1)}, \{h_u^{(l-1)} | u \in \mathcal{N}(v)\} \right) \right)$$

where $\mathcal{N}(v)$ denotes the neighbor nodes of node v ;

- Driving mode: Set a large neighborhood aggregation radius to capture long-distance road network correlations;
- Parking mode: Set a small neighborhood aggregation radius to capture refined parking boundary constraints.

The final output is the global topological embedding feature z_{topo} , which serves as the input of the policy network and value network.

To avoid the computational explosion caused by full-graph encoding of large-scale global road networks, the model adopts a **local topological subgraph perception strategy centered on the ego vehicle**, only encoding the topological areas effective for decision-making to ensure real-time inference.

5.3 AlphaZero-Style Self-Play Core Decision Algorithm

5.3.1 Network Structure Design

1. PolicyNet

Input: Topological embedding feature z_{topo} + ego-vehicle dynamic state features

Output: Prior probability distribution $\pi(a|S)$ of all candidate driving actions, combined with Action Mask to block illegal actions.

2. ValueNet

Input: Consistent with the policy network

Output: Long-term cumulative value estimation $v(S) \in [-1, 1]$ of the current road network game state.

5.3.2 Improved MCTS-UCT Selection Formula (Core Algorithm Formula)

The Upper Confidence Bound for Trees (UCT) is adopted as the MCTS node selection strategy to balance exploration and exploitation:

$$a^* = \arg \max_a \left(\frac{Q(s, a)}{N(s, a)} + c_{puct} \cdot \pi(a|S) \cdot \frac{\sqrt{\sum_b N(s, b)}}{1 + N(s, a)} \right)$$

where:

- $Q(s, a)$: Cumulative reward of node (s, a) ;
- $N(s, a)$: Action visit count;
- c_{puct} : Exploration coefficient controlling the agent's exploration degree of unfamiliar actions;
- $\pi(a|S)$: Prior action probability output by the policy network.

A single MCTS search consists of four steps: **Select** $\rightarrow$ **Expand** $\rightarrow$ **Evaluate** $\rightarrow$ **Backpropagate**.

5.3.3 Single Episode Self-Play Flow Algorithm (Dynamics-Constrained Version)

Algorithm 1: Single Episode Self-Play Flow for Topological Game Autonomous Driving

Input: RoadGenerator, PolicyNet, ValueNet, Maximum episode steps T_{max}
Output: Single episode driving trajectory dataset *trajectory*
Initialize: *road_graph* = RoadGenerator.generate_topo() // Generate modular assembled road game board
current_state = RoadBoardState(*road_graph*) *trajectory* = [] **while** *current_state* is not terminated **and** *steps* < T_{max} **do**
 // Step 1: Encode current topological state via GNN
 state_emb = GNN_Encoder(*current_state*) // Step 2:
 Execute MCTS-UCT multi-step lookahead game search
 root_node = MCTS_Search(*root_state* = *current_state*, *state_emb*, PolicyNet, ValueNet) // Step 3:
 Obtain optimal high-level decision move according to visit frequency
 action_dist = get_action_distribution(*root_node*)
 best_action = sample_action(*action_dist*) // Step 4:
 Record current state and action distribution as training samples
 trajectory.append((current_state, action_dist)) // Step 5:
 Execute move decisions constrained by vehicle bicycle dynamics
 // Generate physically feasible smooth trajectories and update vehicle poses and road game states
 current_state, step_reward = *current_state.step(best_action)*
Obtain the final episode reward *final_value*
Label each sample in *trajectory* with the target value *final_value*
return *trajectory*

Supplementary Explanation of Dynamic Constraints: The discrete moves output by the agent are only high-level sequential decision intentions. The action execution process strictly embeds **vehicle bicycle dynamic constraints** to ensure that each trajectory satisfies physical boundaries such as vehicle speed, steering angle and jerk, avoiding non-physical teleportation and step-like motion, which fully meets the physical feasibility requirements of autonomous driving motion planning.

5.3.4 Network Iterative Update Loss Functions

$$\mathcal{L}_{policy} = - \sum_a \rho_{mcts}(a) \log \pi_{net}(a) \quad (1)$$

$$\mathcal{L}_{value} = (v_{net}(S) - v_{final})^2 \quad (2)$$

$$\mathcal{L}_{total} = \mathcal{L}_{policy} + \lambda \cdot \mathcal{L}_{value} \quad (3)$$

where λ denotes the weight coefficient of value loss.

5.3.5 Policy Distillation Algorithm (Teacher-Student Network)

Algorithm 2: MCTS Teacher Policy Distillation for Lightweight Student Network

Input: MCTS self-play generated teacher trajectory dataset $D_{teacher}$

Output: Lightweight student planning network $StudentNet$

Initialize $StudentNet$ (lightweight MLP / shallow Transformer)

foreach teacher sample ($state_emb, \rho_{mcts}$) **do**

 Student network predicts action distribution $\pi_{student}$

 Calculate distillation loss: $L_{distill} = - \sum \rho_{mcts} \log \pi_{student}$

Optimize $StudentNet$ via gradient descent to fit the optimal decision distribution from MCTS

After distillation, $StudentNet$ can be directly used for real-time vehicle-end planning output

5.4 Dynamic Search Strategy for Different Scenarios

1. **Driving Scenarios:** Set a small MCTS search depth, focusing on short-term lane games and intersection turning decisions;
2. **Parking Scenarios:** Increase MCTS search depth and simulation counts to adapt to long-sequence decision requirements of multi-step parking adjustments and sequential parking entry;
3. Adaptive switching of both scenarios is realized by adjusting the MCTS search depth parameter d_{search} .

6 Differential Advantages of the Proposed Paradigm

Compared with mainstream autonomous driving training paradigms, the proposed topological game framework has five irreplaceable differentiated values:

1. **Reduced Data Dependence and Breaking Computing Competition**
Massive real-world driving trajectories and labeled data are not required. High-quality driving samples are autonomously generated via rule constraints and self-play, adapting to the domestic industrial situation of imperfect data closed loops and scarce high-end computing resources.

2. **Topological Completeness and Provable Scenario Coverage**

Full-domain road networks are generated via combinations of finite primitives, enabling mathematically demonstrable scenario completeness, systematically covering various intersection and road topologies, and thoroughly solving the limitations of random sampling for edge cases.

3. **Interpretable Decision-Making and Safety Controllability**

Abandoning black-box pixel fitting, decisions are deduced based on explicit road topologies, traffic rules and MCTS search trees. The decision logic is traceable, interpretable and verifiable, meeting the safety compliance requirements of high-level autonomous driving.

4. **Strong Topological Generalization for Full-Domain Road Conditions**

The model learns common structural features of road networks rather than scene pixel features, enabling adaptive performance on unfamiliar assembled road networks in different cities and geographies, with far stronger generalization than scene-fitting models.

5. **Full-Domain Unified Modeling and Native Integrated Driving-Parking**

Different from the mainstream “driving model + parking model” dual-architecture stacking scheme in the industry, the proposed framework realizes native integrated decision-making for macro road traversal and micro parking via **unified graph topological priors + unified MCTS game search + unified self-play iteration mechanism**, forming a truly general intelligent driving base at the algorithm level.

Core Difference from Traditional Rule-Based Planning: Different from traditional rule-based motion planning relying on manually designed cost functions, the proposed framework realizes multi-step game foresight via MCTS and implicitly learns multi-agent interaction values through self-play. It can autonomously emerge social driving behaviors that cannot be exhaustively defined by manual rules, such as unprotected left turns, multi-vehicle roundabout merging and parking competition games, showing stronger adaptability to dynamic scenarios and intelligent interaction capabilities.

7 **Core Scientific Assumptions and Risk Verification Scheme**

The effectiveness of the proposed paradigm relies on four core assumptions, which need to be verified and optimized in stages to clarify the framework boundary and improvement directions:

7.1 **Assumption 1: Discrete Sequential Moves Can Represent Safe Driving Strategies**

Core Risk: Discretization of continuous vehicle control may lead to trajectory jitter, decision lag and suboptimal driving performance, failing to match the ride smoothness

of continuous control methods such as Model Predictive Control (MPC).

Verification Scheme: Fix a single intersection scenario without traffic flow, and quantitatively compare the driving efficiency, trajectory smoothness and start-stop jitter indicators between discrete MCTS decisions and continuous MPC control. If defects exist, upgrade to **parameterized trajectory segment moves** to replace single-point instantaneous control, balancing discrete search characteristics and continuous driving smoothness.

7.2 Assumption 2: GNN Has Generalization Ability Across Assembled Topologies

Core Risk: Local neighborhood encoding of GNNs may fail to capture global traffic emergence features brought by multi-primitive assembly, resulting in invalid learned primitive strategies on mixed topology road networks.

Verification Scheme: Adopt separated training and test sets: train only on independent single road primitives, and test on assembled mixed topologies not seen during training. Observe the stability of agent decisions to verify whether the model learns essential topological features rather than scene memorization.

7.3 Assumption 3: Multi-Agent Self-Play Can Emerge Reasonable Social Driving Behaviors

Core Risk: Unconstrained multi-agent self-play may converge to extreme Nash equilibria, leading to over-conservative yielding with low traffic efficiency or aggressive driving with frequent collisions and strategy collapse.

Verification Scheme: Adopt rule-based background agents as behavioral anchors in the early stage to constrain the convergence range of strategies. In the long term, introduce a strategy population mechanism to maintain diverse driving styles and emerge mixed game strategies consistent with real road conditions.

7.4 Assumption 4: Consistency of Cross-Scale Topology Mapping (New for Driving-Parking Extension)

Core Risk: There are huge differences in spatial density, node spacing and feature scales between large-scale driving topologies (hundreds of meters) and small-scale parking topologies (meters). A single GNN may stably learn dual-scale game features with difficulty, leading to scale transfer degradation.

Verification Scheme:

1. Design **multi-scale ablation experiments**: three groups including driving-only training, parking-only training and mixed driving-parking training;
2. Introduce **dynamic aggregation radius multi-scale graph convolution** to adaptively adjust the receptive field of different scenarios;
3. Quantify cross-scene transfer accuracy, decision stability and game success rate to verify the scale generalization of the framework.

7.5 Auxiliary Optimization Risks

Aiming at the subjectivity of manually set reward weights, conduct **reward weight sensitivity scanning experiments** to quantify the influence boundary of weights. In the future, inverse reinforcement learning (IRL) will be introduced to infer optimal reward functions from a small amount of high-quality human driving trajectories and reduce the subjectivity of manual parameter tuning.

8 Phased Experimental R&D Roadmap

To realize steady iteration and complete verification of the framework, a five-stage minimum feasible R&D landing path is formulated, progressing from basic capability verification to full-domain self-play and simulation-to-real transfer:

1. **Basic Verification Stage:** Build a fixed single-intersection topological game board without dynamic traffic flow, verify that the model can autonomously learn basic rules such as signalized traversal, lane keeping and destination arrival;
2. **Single-Agent Interaction Stage:** Introduce IDM rule-based traffic flows and non-motor vehicle agents, verify basic interactive behaviors such as car-following, deceleration and yielding under the hierarchical reward system;
3. **Topological Generalization Verification Stage:** Enable the automatic road network generator to construct road networks assembled from multiple primitives, test the generalization ability of the model on unseen topologies;
4. **Multi-Agent Self-Play Stage:** Deploy multiple homogeneous decision agents, verify the emergence effect of social driving strategies and optimize the balance of game convergence;
5. **Distillation and Domain Transfer Stage:** Complete MCTS policy distillation, connect high-fidelity simulation and real-vehicle intelligent driving pipelines, and realize the transfer of simulation strategies to real-world scenarios.

9 Academic Value and Application Prospects

9.1 Academic Value

This paper breaks the current research status of “emphasizing data, ignoring structure, emphasizing effect, ignoring theory” in autonomous driving, and **first establishes a topological combinatorial game theory system for intelligent driving motion planning**, providing a formally verifiable framework for the scenario completeness, decision generalization and safety interpretability of autonomous driving. This research is not limited to algorithm iteration, but constructs a provable, derivable and generalizable underlying training paradigm for intelligent driving, filling the research gap of topological structure modeling and self-play iteration in the field of intelligent driving.

Relying on the three-layer technical system of **Geometric Deep Learning, Topo-MoE Multi-Scale Topological Encoding, Topology GRPO Game Optimization**, this work systematically solves three core problems: topological representation, cross-scale unification and unstable game training, providing a novel fusion paradigm for the intersection of geometric artificial intelligence and self-play planning.

9.2 Engineering Application Scenarios

1. **Pre-simulation Verification Platform for Autonomous Driving:** Low-cost batch verification of the road network adaptability of various decision algorithms, greatly reducing the cost of real-vehicle testing;
2. **Special Game Decision Module for Complex Intersections:** Targeted solution of high-difficulty interactive scenario decision-making such as roundabouts, unsignalized intersections and ramp merging;
3. **General Decision Training Base for Integrated Driving and Parking**
This framework completely eliminates the algorithm boundary between driving domains and parking domains. Relying on a unified topological game mechanism, unified GNN encoding and unified self-play training logic, it can support full-scenario decision training for highways, urban roads, roundabout ramps, indoor and outdoor parking lots, realizing **truly algorithm-native integrated driving and parking autonomous driving**, which has high industrial landing value compared with traditional hardware-software stacking schemes.

9.3 Future Research Directions

Future work will focus on three iterative directions: optimizing the continuous-adaptive MCTS algorithm to solve the smoothness problem of discrete moves; constructing dynamic topology perception mechanisms to realize real-time understanding and adaptive decision-making of unknown road networks; building an open-source road primitive simulation library to promote the construction of a lightweight intelligent driving research ecosystem.

10 Conclusion

This paper proposes a **topological game-based autonomous driving paradigm**, completely breaking out of the data-computing intensive development path led by Tesla and the traditional industry idea of separated driving and parking design. Seven core road primitives are adopted to realize complete modeling of driving scenarios, and extended parking topological primitives are added to expand the game capability of micro parking fields. Based on unified graph topological representation, unified MCTS game search mechanism and unified self-play iterative training system, this paper first constructs a full-domain general autonomous driving game theoretical framework covering **macro roads + micro parking lots, driving + parking**.

Supported by geometric deep learning, TopoMoE multi-scale expert encoding and Topology GRPO topological game optimization, this paradigm forms a complete closed loop of theory and algorithms. It is not only a novel decision algorithm framework, but also a **formally complete, cross-scene generalizable theoretical system for intelligent driving training and verification**, which can provide a new underlying support for the safe iteration, scenario generalization and full-domain landing of general autonomous driving.

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